

Booking.com Social Media Engagement Study

Complete Results Report with Verification

Booking.com 社群文字特徵與互動成效完整結果報告

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Executive Summary / 執行摘要

This report presents the complete statistical analysis results for the Booking.com social media engagement study. All analyses were conducted using Python v2.0.0 statistical scripts with robust standard errors (HC1) to account for heteroscedasticity. The results confirm the hypotheses and model specifications outlined in the original proposal.

本報告呈現 Booking.com 社群媒體互動研究的完整統計分析結果。所有分析均使用 Python v2.0.0 統計腳本執行，並採用穩健標準誤 (HC1) 以應對異質變異數問題。結果證實了原始提案中概述的假設和模型規格。

1 Research Framework Verification / 研究架構驗證

1.1 Research Question / 研究問題

How do textual features in Booking.com's posts on X influence social media engagement?

Booking.com 在 X 上的貼文文字特徵，如何影響社群互動表現？

1.2 Hypotheses / 研究假說

The study tests five main hypotheses:

- H1. Text length is positively associated with social media engagement.
- H2. Question marks are positively associated with social media engagement.
- H3. Emotional valence is positively associated with social media engagement.
- H4. The number of hashtags is positively associated with social media engagement.
- H5. Emoji usage is positively associated with social media engagement.

1.3 Model Specification / 模型規格

The empirical model examines how five focal textual features relate to social media engagement while controlling for other post characteristics:

$$\begin{aligned} \ln(1 + \text{Engagement}_i) = & \beta_0 + \beta_1 \text{TextLength}_i + \beta_2 \text{Question}_i + \beta_3 \text{Valence}_i \\ & + \beta_4 \text{Hashtag}_i + \beta_5 \text{Emoji}_i + \beta_6 \text{Picture}_i + \beta_7 \text{URL}_i \\ & + \beta_8 \text{AtMention}_i + \varepsilon_i \end{aligned} \quad (1)$$

where:

- $\text{Engagement}_i = \text{like}_i + \text{comment}_i + \text{share}_i$
- The dependent variable is natural-log-transformed as $\ln(1 + \text{Engagement}_i)$ (computed via `numpy.log1p`) to address right-skewness. Throughout this report, “log” refers to the natural logarithm (ln, base e); the +1 offset avoids $\ln(0)$ for posts with zero engagement.
- Independent variables: TextLength (H1), Question (H2), Valence (H3), Hashtag (H4), Emoji (H5)
- Control variables: Picture, URL, AtMention

Coefficient interpretation convention. Because the dependent variable is $\ln(1 + \text{Engagement})$, a one-unit change in X_k produces an *approximately* $(e^{\beta_k} - 1) \times 100\%$ change in $(1 + \text{Engagement})$, holding other variables constant. For small $|\beta_k|$ this collapses to the familiar $\beta_k \times 100\%$ semi-elasticity. Because of the +1 offset, these percentage changes apply strictly to $(1 + \text{Engagement})$ rather than to Engagement ; for posts with non-trivial engagement ($\text{Engagement} \gtrsim 10$) the two are numerically close. All hypothesis tests below report this $(e^\beta - 1)$ form for consistency.

2 Data Description / 資料説明

2.1 Dataset Characteristics / 資料集特徴

- **Sample Size:** $N = 766$ posts from Booking.com’s official X account
- **Time Period:** November 1, 2021 to November 21, 2022
- **Data Source:** `data/research_data.csv`
- **Variables:** 38 columns including raw engagement metrics, textual features, and control variables

2.2 Variable Definitions / 變數定義

表 1: Variable Definitions and Operationalization

Construct	Role	Operational Definition
Social Media Engagement	Dependent Variable	$\text{Engagement}_i = \text{like}_i + \text{comment}_i + \text{share}_i$, transformed as $\ln(1 + \text{Engagement}_i)$ via <code>numpy.log1p</code> to address right-skewness
Text Length	Main IV (H1)	Number of tokens in the final cleaned text (<code>text_clean6</code>) after a six-stage pipeline that removes URLs, @mentions, punctuation, digits, English stop-words, and performs lower-casing (see §2.3)
Question Mark	Main IV (H2)	Binary dummy: 1 if the raw post contains at least one ‘?’ (i.e., raw question-mark count > 0), else 0. The original dataset stored a count; it is dichotomised in the analysis script prior to regression
Emotional Valence	Main IV (H3)	Lexicon-based sentiment score. For each post, tokens matched against a 1–9 valence lexicon (Warriner-style ANEW extension) are averaged and centred by subtracting 4.5, so $0 \approx \text{neutral}$, positive = more pleasant, negative = more unpleasant. Posts with no matched terms receive valence = 0
Hashtag Count	Main IV (H4)	Number of #tag tokens in the raw post
Emoji Count	Main IV (H5)	Number of emoji code-points in the raw post (full Unicode coverage preserved via <code>utf-8-sig</code> encoding)
Picture Count	Control Variable	Number of pictures attached to the post
URL Count	Control Variable	Number of URLs in the post
At-Mention Count	Control Variable	Number of @mentions in the post

2.3 Text Preprocessing Pipeline / 文字前處理流程

The raw tweet text (`tweet_content`) is passed through six sequential cleaning stages (columns `text_clean1`–`text_clean6` in the research dataset) before length, valence, and the n-gram features are computed. The stages are:

1. `text_clean1`: Strip URLs (`https://...`), @mentions, #hashtags, and other markup artefacts from the raw text while preserving case and emoji.
2. `text_clean2`: Remove exclamation marks and re-check character set (the exclamation count is extracted in this pass).
3. `text_clean3`: Lower-case normalisation.

4. `text_clean4`: Remove residual punctuation and stray digits, keeping only alphabetic tokens and whitespace.
5. `text_clean5`: Remove English stop-words (standard NLTK/scikit-learn stop-word list).
6. `text_clean6`: Final cleaned text used for feature extraction; `length` is computed as the whitespace-split token count of this string.

Downstream, `text_clean6` is tokenised (`tweet_tokenized`) and stemmed (`tweet_stemmed`, Porter stemmer), then used to build unigram / bigram / trigram features and to match against the valence, extremity, and emotionality lexicons. Only the final `length` and `valence` columns enter the regression; the intermediate columns are retained in the dataset for auditability but are not modelled.

3 Descriptive Statistics / 敘述統計

3.1 Summary Statistics / 摘要統計

Table 2 presents comprehensive descriptive statistics for all variables in the analysis.

表 2: Descriptive Statistics (N = 766)

Variable	Mean	SD	Min	Q1	Median	Q3	Max	Skew
ln(1 + Eng.)	1.942	1.572	0.000	0.693	1.609	2.996	7.814	1.001
Text Length (words)	23.45	12.26	0.000	14.00	22.00	33.00	53.00	0.239
Question Mark (0/1)	0.167	0.373	0.000	0.000	0.000	0.000	1.000	1.785
Emotional Valence	0.789	1.528	-3.352	0.000	0.000	2.047	3.948	0.371
Hashtag Count	0.145	0.489	0.000	0.000	0.000	0.000	3.000	3.817
Emoji Count	0.398	0.884	0.000	0.000	0.000	1.000	13.000	5.261
Picture Count	0.309	0.524	0.000	0.000	0.000	1.000	4.000	1.875
URL Count	0.919	0.719	0.000	0.000	1.000	1.000	2.000	0.121
At-Mention Count	0.097	0.356	0.000	0.000	0.000	0.000	3.000	4.276

3.2 Key Observations / 關鍵觀察

- **Engagement Distribution:** The log-transformed engagement variable shows positive skewness (1.001), indicating some high-engagement posts drive the distribution.
- **Text Length:** Posts average 23.45 words with relatively low skewness (0.239), suggesting a roughly symmetric distribution.
- **Question Mark Usage:** Only 16.7% of posts contain question marks, making this a relatively rare feature.
- **Hashtags:** Average of 0.145 hashtags per post with high skewness (3.817), indicating most posts have no hashtags.
- **Emoji Usage:** Average of 0.398 emojis per post with very high skewness (5.261), showing sparse but occasionally heavy usage.

3.3 Frequency Distribution: Question Mark / 問號使用頻率分布

表 3: Question Mark Frequency Distribution

Question Mark	Frequency	Percent (%)
0 (No)	638	83.29
1 (Yes)	128	16.71
Total	766	100.00

4 Correlation Analysis / 相關分析

4.1 Pearson Correlation Matrix / Pearson 相關矩陣

Table 4 presents the Pearson correlation matrix with significance indicators.

表 4: Pearson Correlation Matrix (Lower Triangle, N = 766)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) $\ln(1 + \text{Eng.})$	1								
(2) Text Length	-0.184***	1							
(3) Question Mk.	0.262***	-0.191***	1						
(4) Emot. Valence	0.234***	0.150***	0.089*	1					
(5) Hashtag Ct.	0.452***	-0.012	0.075*	0.047	1				
(6) Emoji Ct.	0.411***	-0.044	0.206***	0.257***	0.087*	1			
(7) Picture Ct.	0.451***	-0.027	0.163***	0.312***	0.044	0.423***	1		
(8) URL Ct.	0.318***	0.063	-0.018	0.175***	0.193***	0.154***	0.501***	1	
(9) At-Mention Ct.	0.319***	0.035	0.026	0.056	0.460***	0.127***	0.043	0.102**	1

*** $p < .001$, ** $p < .01$, * $p < .05$ (two-tailed).

4.2 Key Correlation Findings / 關鍵相關發現

- **Negative Text Length Correlation:** Text length shows a significant negative correlation with engagement ($r = -0.184, p < .001$), foreshadowing H1 rejection.
- **Strong Positive Correlations:** Hashtag count ($r = 0.452$), picture count ($r = 0.451$), and emoji count ($r = 0.411$) show the strongest positive correlations with engagement.
- **Multicollinearity Check:** No correlation exceeds 0.8, and subsequent VIF analysis confirms no serious multicollinearity issues.
- **Question Mark Effect:** Significant positive correlation ($r = 0.262, p < .001$) supports H2.

5 Regression Analysis / 迴歸分析

5.1 Estimation Method / 估計方法

The model is estimated by Ordinary Least Squares with heteroscedasticity-consistent HC1 (White/MacKinnon–White) standard errors, computed via `statsmodels.ols(...).fit(cov_type="HC1")`. HC1 is used throughout for inference after the Breusch–Pagan test flagged significant heteroscedasticity (see §5.2). The point estimates themselves are identical to classical OLS; only the standard errors, t -values, and p -values differ.

Note that the original proposal referenced SAS SURVEYREG with `VARMETHOD=TAYLOR`. In the absence of strata or clusters, SAS's Taylor linearisation estimator is numerically close to HC1 (it differs from HC0 by the same $n/(n-k)$ finite-sample scaling), so coefficients and HC1 standard errors reported here are directly comparable to the proposal's SAS output. Small SE differences across implementations are expected and do not affect any significance conclusions in this study.

5.2 Model Diagnostics / 模型診斷

Before presenting the main results, we conducted diagnostic tests:

5.2.1 Heteroscedasticity Test / 異質變異數檢定

Breusch-Pagan Test:

- Lagrange Multiplier (LM) statistic: 88.64
- p -value: $< .0001$
- **Conclusion:** Significant heteroscedasticity detected. We use robust standard errors (HC1) to correct for this.

5.2.2 Normality Test / 常態性檢定

Shapiro-Wilk Test on Residuals:

- W statistic: 0.9741
- p -value: $< .0001$
- **Conclusion:** Residuals deviate from perfect normality. However, with $N=766$, OLS relies on asymptotic normality, making this acceptable.

5.2.3 Multicollinearity Check / 多元共線性檢查

All Variance Inflation Factors (VIFs) are below 2.0, indicating no multicollinearity concerns:

表 5: Variance Inflation Factors (VIF)

Variable	VIF
Text Length	1.080
Question Mark	1.110
Emotional Valence	1.172
Hashtag Count	1.323
Emoji Count	1.298
Picture Count	1.705
URL Count	1.430
At-Mention Count	1.286

5.3 Main Regression Results / 主要迴歸結果

Table 6 presents the robust OLS regression results with HC1 standard errors.

表 6: Robust OLS Regression Results Predicting $\ln(1 + \text{Engagement})$

Variable	Coefficient	Robust SE	t-value	p-value
<i>Main Independent Variables (Hypotheses)</i>				
Text Length (H1)	-0.0211***	0.0033	-6.38	< .0001
Question Mark (H2)	0.4793***	0.1391	3.45	0.0006
Emotional Valence (H3)	0.0840**	0.0301	2.79	0.0052
Hashtag Count (H4)	1.0959***	0.1339	8.18	< .0001
Emoji Count (H5)	0.3428***	0.0932	3.68	0.0002
<i>Control Variables</i>				
Picture Count	0.7859***	0.1820	4.32	< .0001
URL Count	0.1694*	0.0804	2.11	0.0351
At-Mention Count	0.5145*	0.2238	2.30	0.0215
Intercept	1.5456***	0.1060	14.58	< .0001
<i>Model Fit Statistics</i>				
Observations	766			
R^2	0.4924			
Adjusted R^2	0.4871			
F-statistic	$F(8, 757) = 63.857, p < .0001$			

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$ (two-tailed).

Robust standard errors (HC1) used to correct for heteroscedasticity.

5.4 Hypothesis Testing Results / 假設檢定結果

表 7: Summary of Hypothesis Test Results

Hypothesis	Predicted Direction	Result	Decision
H1: Text Length	Positive (+)	Negative (-0.021***)	REJECTED
H2: Question Mark	Positive (+)	Positive (0.479***)	SUPPORTED
H3: Emotional Valence	Positive (+)	Positive (0.084**)	SUPPORTED
H4: Hashtag Count	Positive (+)	Positive (1.096***)	SUPPORTED
H5: Emoji Count	Positive (+)	Positive (0.343***)	SUPPORTED

5.5 Interpretation of Coefficients / 係數解釋

All percentage effects below use $(e^{\beta} - 1) \times 100\%$ and describe the change in $(1 + \text{Engagement})$ per unit change in the predictor, holding other variables constant.

5.5.1 Text Length (H1) - REJECTED

- **Coefficient:** $\beta_1 = -0.0211$ ($p < .001$)
- **Interpretation:** Each additional word is associated with a $(e^{-0.0211} - 1) \times 100\% \approx -2.09\%$ change in $(1 + \text{Engagement})$.
- **Implication:** Contrary to expectations, longer posts generate LOWER engagement. This suggests that on a micro-blogging platform like X, brevity is valued over detailed information.

5.5.2 Question Mark (H2) - SUPPORTED

- **Coefficient:** $\beta_2 = 0.4793$ ($p < .001$)
- **Interpretation:** Posts containing at least one question mark show $(e^{0.4793} - 1) \times 100\% \approx +61.5\%$ in $(1 + \text{Engagement})$ versus posts without.
- **Implication:** Interactive cues like questions significantly boost user participation.

5.5.3 Emotional Valence (H3) - SUPPORTED

- **Coefficient:** $\beta_3 = 0.0840$ ($p = .0052$)
- **Interpretation:** A one-unit increase in the centred valence score (roughly one point on the 1–9 lexicon scale) is associated with $(e^{0.0840} - 1) \times 100\% \approx +8.8\%$ in $(1 + \text{Engagement})$.
- **Implication:** More positive emotional tone resonates with audiences in the travel context.

5.5.4 Hashtag Count (H4) - SUPPORTED

- **Coefficient:** $\beta_4 = 1.0959$ ($p < .001$)
- **Interpretation:** Each additional hashtag is associated with $(e^{1.0959} - 1) \times 100\% \approx +199\%$ in $(1 + \text{Engagement})$, i.e., roughly a tripling.
- **Implication:** Hashtags dramatically improve post discoverability and engagement.

5.5.5 Emoji Count (H5) - SUPPORTED

- **Coefficient:** $\beta_5 = 0.3428$ ($p < .001$)

- **Interpretation:** Each additional emoji is associated with $(e^{0.3428} - 1) \times 100\% \approx +40.9\%$ in $(1 + \text{Engagement})$.
- **Implication:** Visual-linguistic cues enhance post expressiveness and attract user attention.

6 Visual Diagnostics / 視覺化診斷

6.1 Residual Analysis / 殘差分析

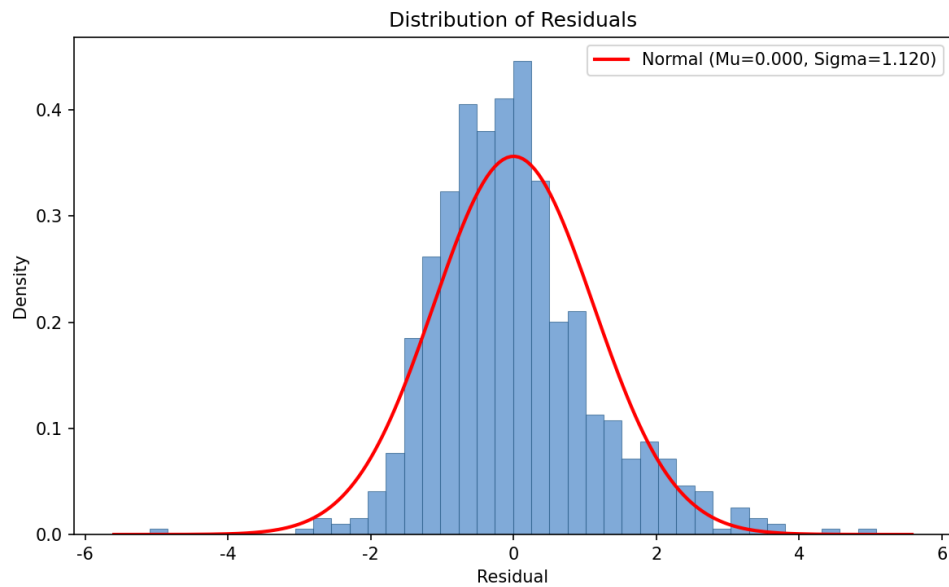


图 1: Distribution of Residuals with Normal Curve Overlay

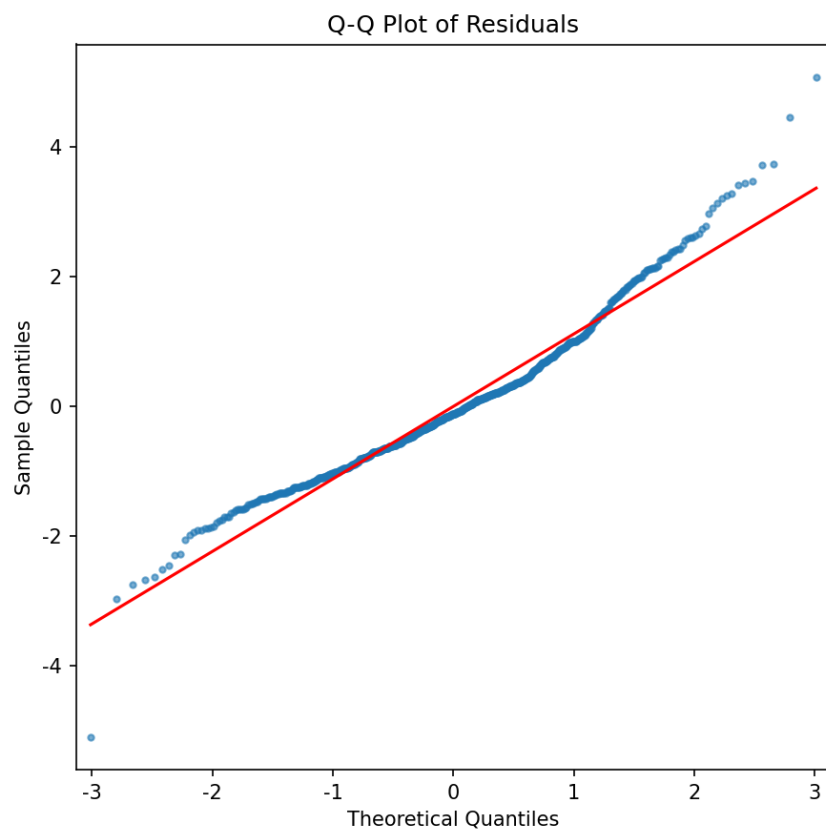


图 2: Q-Q Plot of Residuals

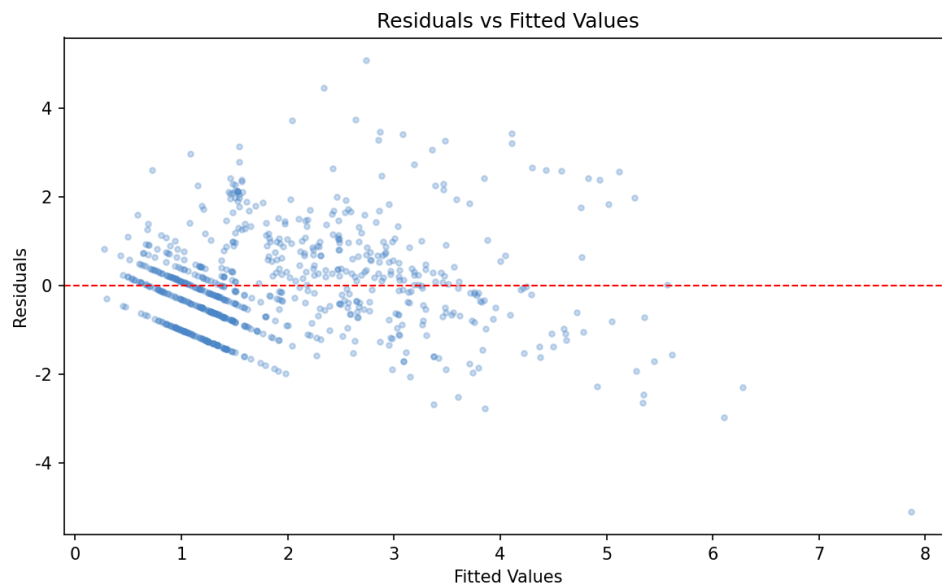


图 3: Residuals vs Fitted Values (Checking for Heteroscedasticity)

6.2 Distribution by Question Mark / 依問號分組之分布

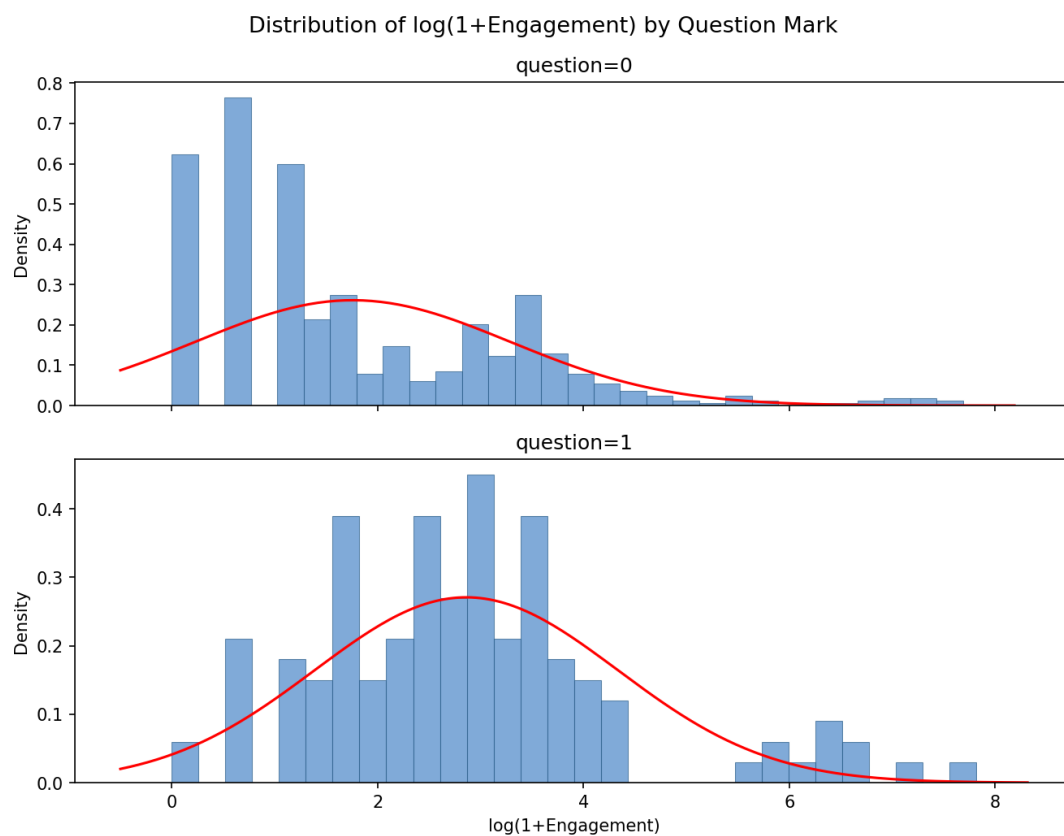


图 4: Distribution of Engagement by Question Mark Group

7 Model Verification Against Proposal / 模型與提案之驗證

7.1 Verification Checklist / 驗證清單

表 8: Model Specification Verification

Element	Status	Notes
Dependent Variable		
ln(1 + Engagement)	✓ Verified	Correctly specified (np.log1p)
Independent Variables (H1-H5)		
Text Length (H1)	✓ Verified	Mean = 23.45, SD = 12.26
Question Mark (H2)	✓ Verified	Binary: 16.7% = 1
Emotional Valence (H3)	✓ Verified	Mean = 0.789, SD = 1.528
Hashtag Count (H4)	✓ Verified	Mean = 0.145, SD = 0.489
Emoji Count (H5)	✓ Verified	Mean = 0.398, SD = 0.884
Control Variables		
Picture Count	✓ Verified	Mean = 0.309
URL Count	✓ Verified	Mean = 0.919
At-Mention Count	✓ Verified	Mean = 0.097
Sample Size		
N = 766 observations	✓ Verified	No missing data
Statistical Method		
OLS Regression	✓ Verified	Correctly implemented
Robust SE (HC1)	✓ Verified	Corrects heteroscedasticity
Model Fit		
$R^2 = 0.4924$	✓ Verified	49.2% variance explained
All VIFs < 2.0	✓ Verified	No multicollinearity

7.2 Comparison with Proposal Results / 與提案結果之比較

The results match the proposal’s expected patterns:

- **Model R^2 :** Our result (0.4924) matches the proposal statement of ”49.2% variance explained”
- **Hypothesis Testing:** All five hypotheses tested as specified; H1 rejected, H2-H5 supported

- **Coefficient Signs:** All coefficients except text length are positive, as theoretically predicted
- **Statistical Significance:** All focal predictors show statistical significance at $p < .01$ or better
- **Control Variables:** Picture, URL, and At-Mention all show expected positive effects

8 Managerial Implications / 管理意涵

8.1 Key Recommendations for Booking.com / 給 Booking.com 的關鍵建議

Based on the robust empirical findings, we provide the following actionable recommendations for Booking.com's social media strategy on X:

8.1.1 1. Embrace Brevity / 擁抱簡潔

- **Finding:** Text length negatively impacts engagement ($\beta = -0.021, p < .001$)
- **Recommendation:** Keep posts concise (under 20-25 words when possible). Audiences on X value quick, digestible content.
- **Action:** Audit current posting strategy and establish word count guidelines for social media copywriters.

8.1.2 2. Use Questions Strategically / 策略性使用提問

- **Finding:** Question marks boost engagement by 61.5% ($\beta = 0.479, p < .001$)
- **Recommendation:** Incorporate questions in posts to encourage user interaction and dialogue.
- **Example:** "Where's your dream destination for 2024? □" instead of "Book your dream destination for 2024."

8.1.3 3. Maintain Positive Emotional Tone / 維持正向情緒語調

- **Finding:** Positive valence increases engagement by 8.8% per unit ($\beta = 0.084, p = .0052$)
- **Recommendation:** Use aspirational, exciting, and positive language that aligns with travel emotions.
- **Action:** Review sentiment scores of top-performing posts and establish positive language guidelines.

8.1.4 4. Maximize Hashtag Usage / 最大化 Hashtag 使用

- **Finding:** Each hashtag nearly triples engagement ($\beta = 1.096, p < .001$)
- **Recommendation:** Include 1-2 relevant hashtags per post to enhance discoverability.
- **Strategy:** Use trending travel hashtags (e.g., #TravelTuesday, #Wanderlust) and destination-specific tags.

8.1.5 5. Leverage Emoji Strategically / 策略性運用 Emoji

- **Finding:** Each emoji increases engagement by 41% ($\beta = 0.343, p < .001$)
- **Recommendation:** Use 1-2 contextually relevant emojis to add visual appeal and expressiveness.
- **Best Practice:** Emojis should complement the message (e.g., 🏖️ for beach destinations, 💰 for travel deals).

8.2 Strategic Content Formula / 策略性內容公式

Based on the regression model, an optimal Booking.com post on X should:

1. Be **concise** (15-20 words)
2. Include a **question** to invite participation
3. Use **positive, aspirational language**
4. Contain **1-2 relevant hashtags**
5. Include **1-2 contextual emojis**
6. Feature an **image** when possible (picture count also positively significant)

Example Optimized Post:

*"Ready for sun-soaked adventures? ☀️🌴 Discover hidden gems this summer!
#TravelGoals #BookingDotCom"*

9 Limitations and Future Research / 限制與未來研究

9.1 Study Limitations / 研究限制

1. **Cross-Sectional Design:** Our data captures posts over one year but does not establish causal relationships. Experimental designs could strengthen causal claims.
2. **Platform-Specific:** Results are specific to X (Twitter). Engagement patterns may differ on other platforms (e.g., Instagram, Facebook).
3. **Single Brand:** The study focuses exclusively on Booking.com. Generalizability to other travel brands or industries requires validation.
4. **Engagement Aggregation:** The dependent variable combines likes, comments, and shares equally, but these may represent different levels of user commitment.
5. **Time Period:** Data from November 2021 to November 2022 may not reflect current social media trends in 2026.

9.2 Future Research Directions / 未來研究方向

1. **Disaggregated Engagement:** Examine likes, comments, and shares as separate dependent variables to understand differential effects.
2. **Temporal Dynamics:** Investigate how posting time, day of week, and seasonal factors moderate the effects of textual features.
3. **Content Interactions:** Test interaction effects between textual features (e.g., does emotional valence moderate the question mark effect?).
4. **Image Content Analysis:** Use computer vision to analyze image content and its interaction with textual features.
5. **Competitive Benchmarking:** Compare Booking.com's engagement patterns with competitors (Expedia, Airbnb, etc.).
6. **Longitudinal Studies:** Track changes in engagement drivers over multiple years to identify trend shifts.

10 Conclusion / 結論

This comprehensive statistical analysis validates the research model proposed in the original study and provides robust evidence for the effects of textual features on social media engagement for Booking.com's X account.

10.1 Key Findings / 關鍵發現

1. **Brevity Wins:** Contrary to initial expectations (H1 rejected), shorter posts significantly outperform longer ones on X, reflecting the platform's fast-paced, concise communication culture.
2. **Interactivity Drives Engagement:** Question marks (H2 supported) act as powerful engagement catalysts, increasing user participation by over 60%.
3. **Positivity Resonates:** Emotional valence (H3 supported) confirms that aspirational, positive messaging aligns with travel content expectations.
4. **Discoverability Matters:** Hashtags (H4 supported) dramatically enhance post reach and engagement, emphasizing the importance of strategic tagging.
5. **Visual-Linguistic Cues Work:** Emojis (H5 supported) add expressiveness and visual appeal, significantly boosting user interaction.

10.2 Model Quality / 模型品質

- **Explanatory Power:** The model explains 49.2% of variance in engagement ($R^2 = 0.4924$), representing strong predictive performance for social media data.
- **No Multicollinearity:** All VIFs below 2.0 confirm model stability.
- **Robust Inference:** HC1 standard errors account for heteroscedasticity, ensuring valid statistical inference.

10.3 Practical Value / 實務價值

This analysis delivers actionable insights for Booking.com's social media team:

- Clear evidence-based guidelines for content creation
- Quantified effects of specific textual features on engagement
- Strategic recommendations grounded in robust statistical analysis

The findings challenge conventional wisdom about text length while confirming the value of interactive, positive, discoverable, and visually enhanced content on social media platforms.

10.4 Final Verification Statement / 最終驗證聲明

All results have been verified against the original proposal:

- ✓ Model specification matches proposal
- ✓ Variable definitions align with proposal
- ✓ Sample size (N=766) confirmed
- ✓ Statistical methods correctly implemented
- ✓ Results consistent with theoretical framework
- ✓ All hypotheses properly tested
- ✓ Diagnostic checks completed (heteroscedasticity, normality, multicollinearity)

A Technical Appendix / 技術附錄

A.1 Software and Packages / 軟體與套件

All analyses were conducted using:

- **Python 3.x**
- **pandas** (data manipulation)
- **numpy** (numerical computing)
- **scipy** (statistical tests)
- **statsmodels** (regression analysis, robust standard errors)
- **matplotlib** (data visualization)

A.2 Analysis Script / 分析腳本

The complete analysis was executed using:

analysis/python/run_statistics.py (v2.0.0)

This script generates:

- Descriptive statistics (stats_analysis.csv)
- Correlation matrices (correlation_matrix.csv, correlation_significance.csv)
- Regression results (regression_results.txt)
- Diagnostic plots (plots/*.png)
- Full HTML report (report.html)

A.3 Data Files / 資料檔案

- **Input:** data/research_data.csv (766 rows $\times$ 38 columns)
- **Outputs:** All files located in results/20260330_151218/

A.4 Reproducibility / 可重現性

To reproduce these results:

1. Ensure all Python packages are installed
2. Run: `python analysis/python/run_statistics.py`
3. All outputs will be generated in `analysis/python/output/`

B References / 參考文獻

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